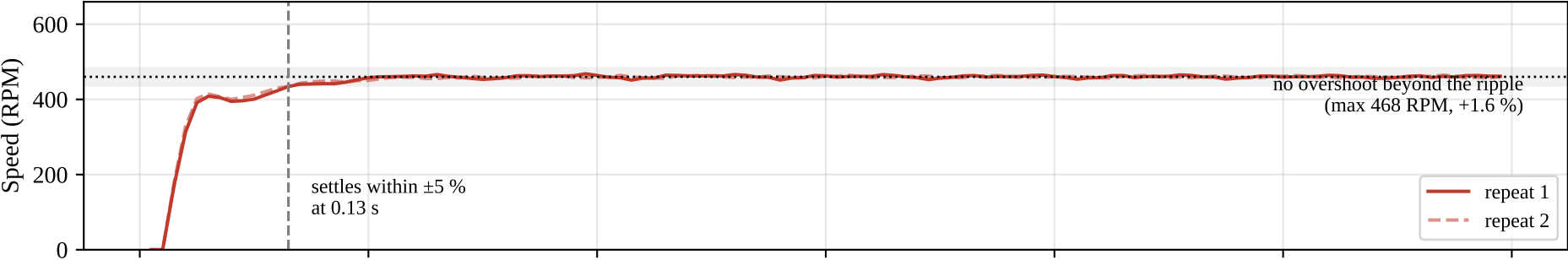
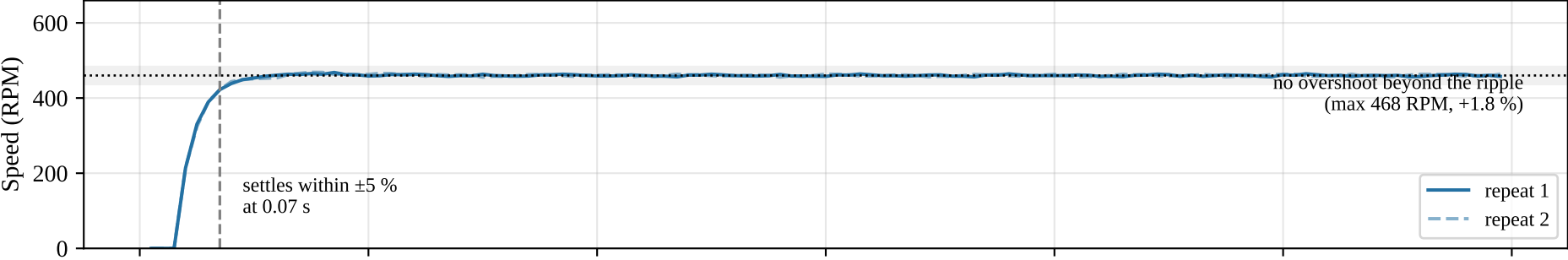


Step response 0 → 460 RPM, one panel per setting (solid = repeat 1, dashed = repeat 2; grey band = ± 5 %)

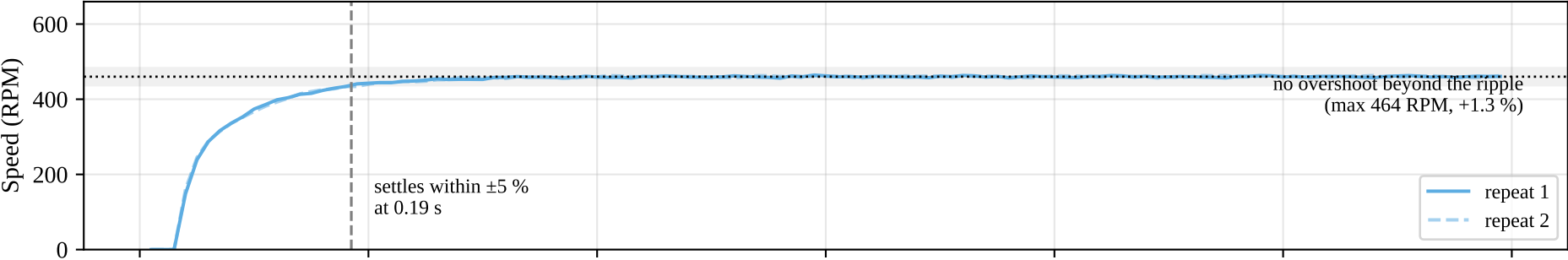
A: baseline (Kp 1.56, Ki 41.2) → overshoot 1.6 %, settles in 0.13 s



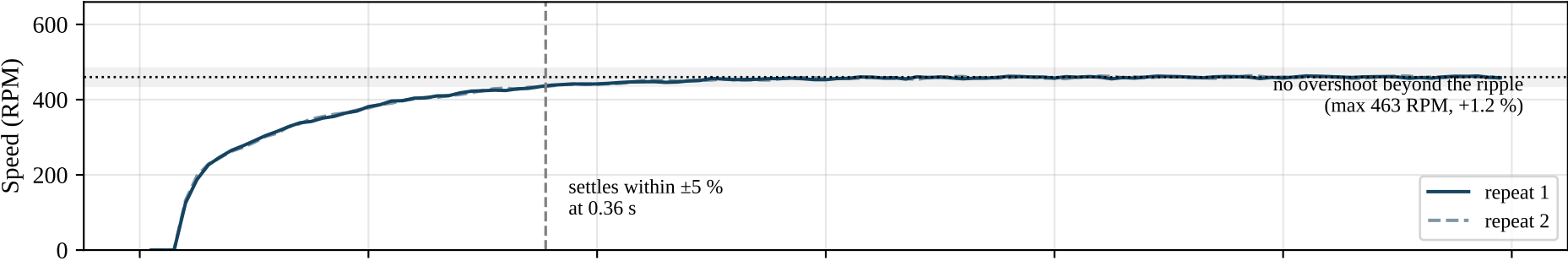
C3: Kp 0.8 → overshoot 1.8 %, settles in 0.07 s



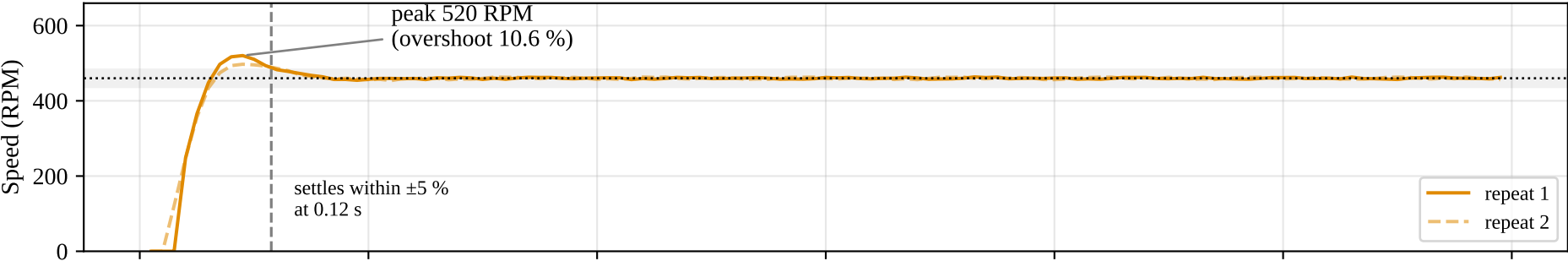
F2: Kp 0.8, Ki 25.0 → overshoot 1.3 %, settles in 0.19 s



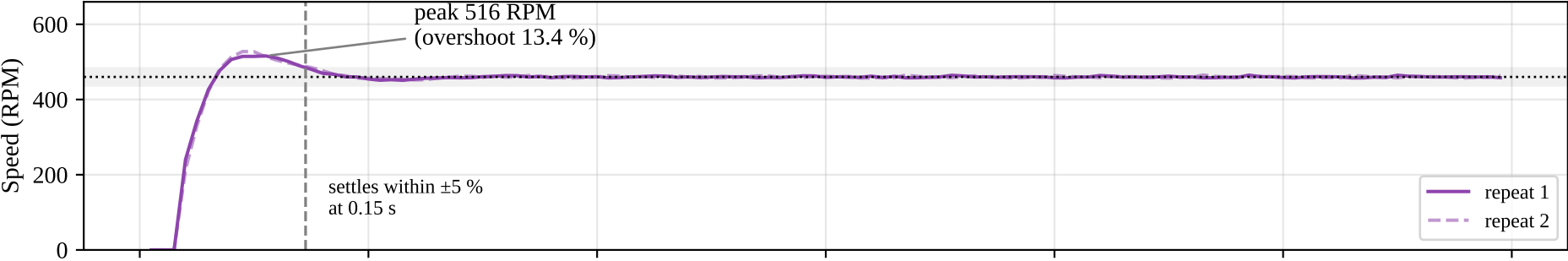
F4: Kp 0.8, Ki 15.0 → overshoot 1.2 %, settles in 0.36 s



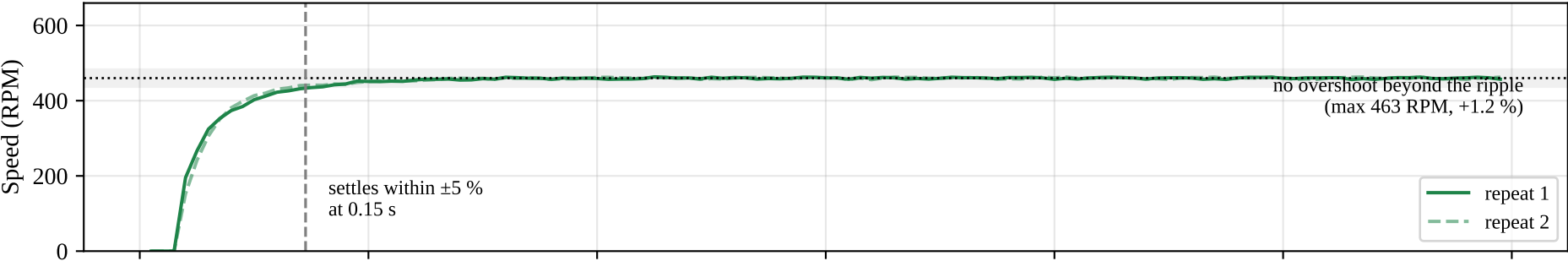
D3: Kp 1.0, α 0.5 → overshoot 10.6 %, settles in 0.12 s



F1: Kp 0.8, α 0.5 → overshoot 13.4 %, settles in 0.15 s



F3: Kp 0.8, Ki 25.0, α 0.5 → overshoot 1.2 %, settles in 0.15 s



Time (s)